PRODUCT CASE STUDY · 2026

Bechthold Industries

LOCAL-FIRST MARKET ANALYSIS

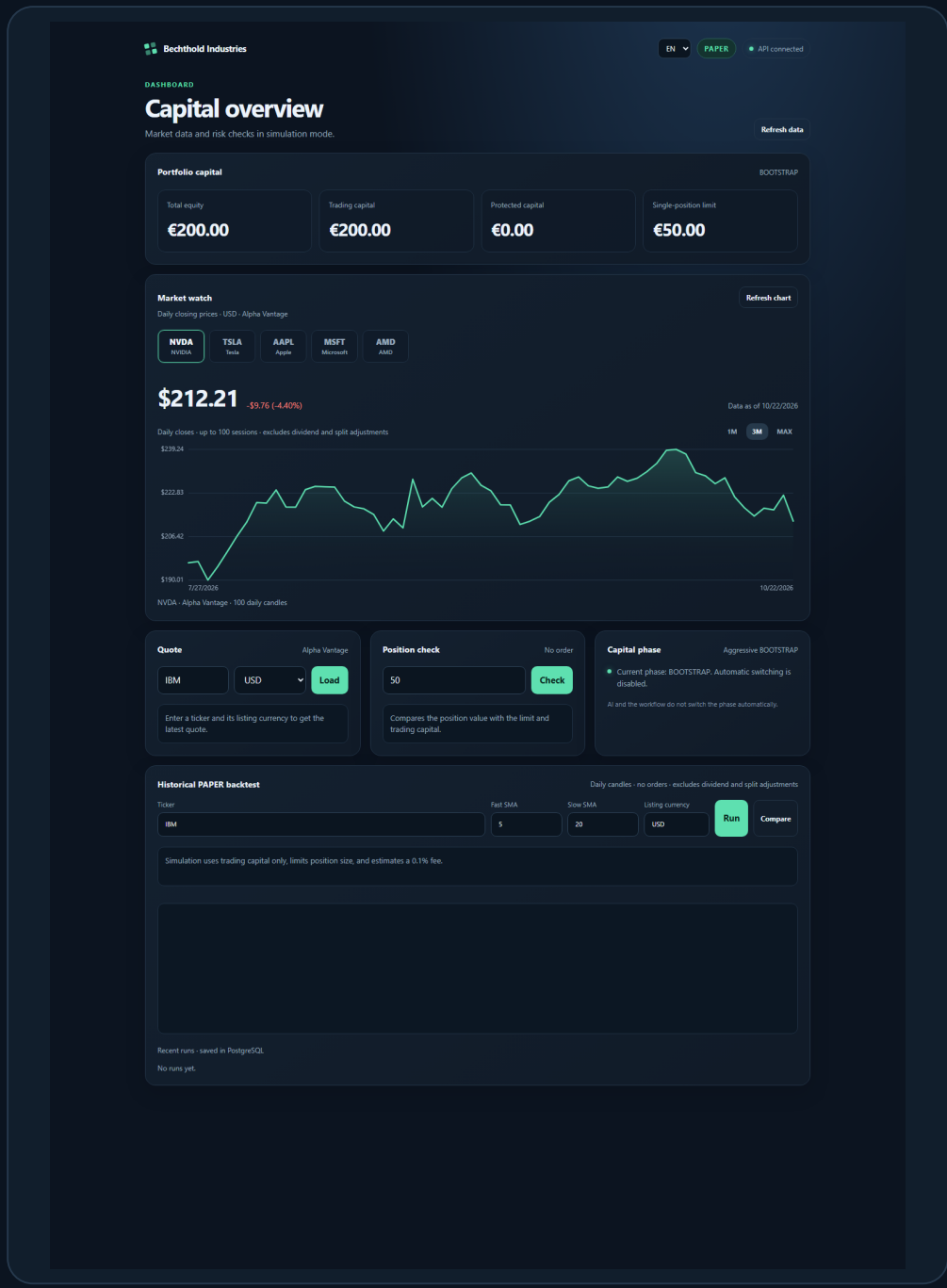
AI Market Platform

A responsive desktop and iPhone dashboard for stock monitoring, capital protection and repeatable strategy research.

€200	€50	5
starting equity	max position	curated tickers

BUILT AROUND CLEAR CAPITAL BOUNDARIES

- Total equity, trading capital and protected capital stay separate.
- BOOTSTRAP → PROFIT_ONLY requires manual confirmation.
- PAPER simulation only; no broker connection or live orders.



Market overview, at a glance

A compact watchlist selector opens an interactive daily price chart. Capital and risk controls remain visible beside the research tools.

MARKET WATCH

Switch between NVDA, TSLA, AAPL, MSFT and AMD. Inspect 1M, 3M or maximum available daily history.

CAPITAL CLARITY

Track total equity, trading capital, protected capital and the per-position limit as separate values.

RISK GATE

Validate a proposed position against both the €50 limit and currently available trading capital.

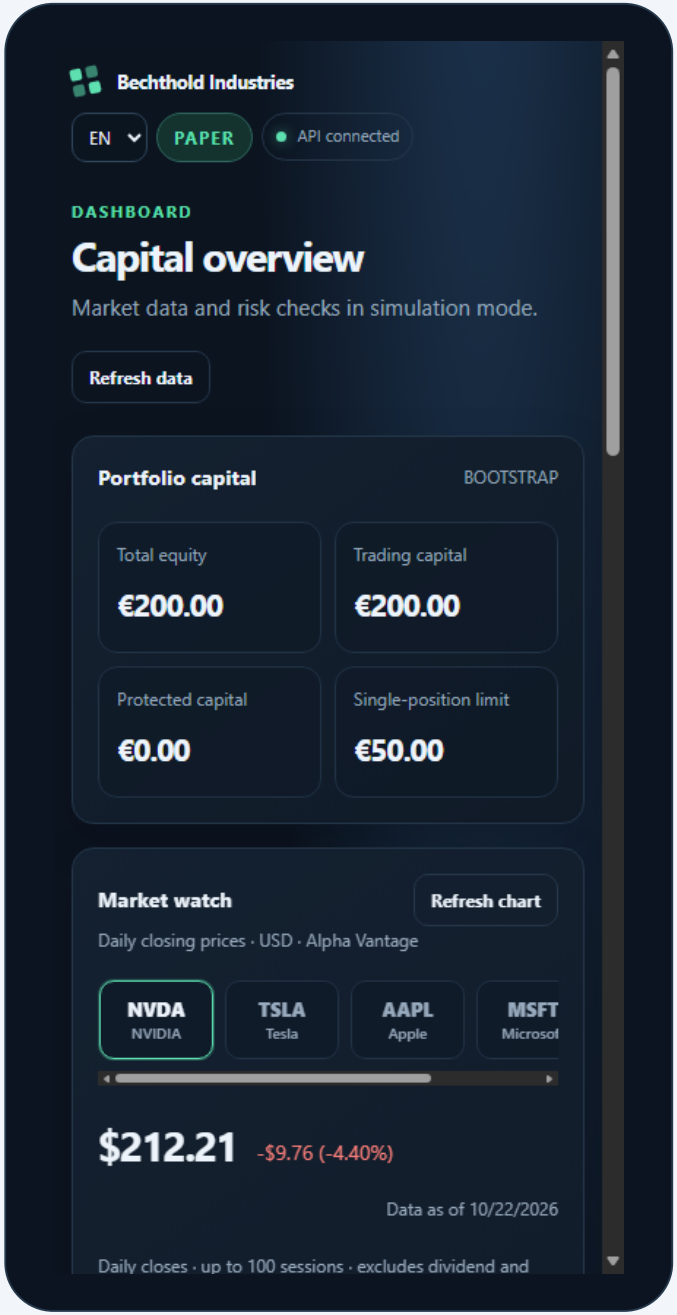
PAPER RESEARCH

Run SMA crossover backtests and a small train/holdout comparison. Results are analytical only.

INSTALLABLE LOCAL WEB APP

Open the dashboard on a computer and add it to an iPhone home screen through Safari while both devices share a private Wi-Fi network. The computer hosts the app; no public deployment is required.

Market data: Alpha Vantage · Currency: listing currency · History: up to 100 daily closes



A deliberately bounded system

A self-hosted tool for one user, with risk rules enforced by the API and an explicit PAPER-only boundary.

DESKTOP / IPHONE
Responsive PWA

→

FASTAPI
Dashboard · risk · analytics

→

POSTGRESQL
Capital · cache · history

→

ALPHA VANTAGE
Quotes · daily prices

N8N PAPER WORKFLOW

Market data → Strategy Engine → risk gate → Position Sizing → final result fields. No automatic phase changes and no broker execution.

CAPITAL PROTECTION

At €350 total equity, €200 is reserved. The BOOTSTRAP → PROFIT_ONLY transition remains pending until the user confirms it with a protected API token. Protected capital is excluded from every risk and sizing calculation.

At €1,000+, the risk profile becomes CONSERVATIVE. Capital state is persisted in PostgreSQL.

VERIFICATION

43	API regression tests passing
7	n8n workflow tests passing
GREEN	GitHub Actions on main

KNOWN LIMITS

Daily prices are raw and may omit split/dividend adjustments; provider quotas can restrict refreshes. Backtests are simplified and do not model slippage, taxes or all market effects. The app is a personal local tool, not a hardened multi-user service or a native iOS app.